Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*
Code: 100310 *Subject: Engineering Mechanics* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

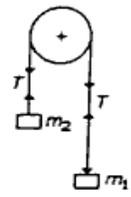
Q.1 ***Choose the correct answer of the following (any seven question only):*** **[2 x 7 = 14]**

(a) $C1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is an identity matrix then it is equivalent to perform rotation
(i) 3 (ii) 1 (iii) 2 (iv) 0

(b) Two cylinders have the same mass and radius. One is hollow and the other is solid. Which one will have the greater moment of inertia about the central axis?
(i) Hollow cylinder (ii) Solid cylinder
(iii) Same for both (iv) Depends on length of cylinder

(c) Single force and a couple acting in the same plane upon a rigid body
(i) balance each other (ii) cannot balance each other
(iii) produce moment of a couple (iv) are equivalent

(d) If the masses of both the bodies, as shown in the figure, are doubled, then the acceleration in the string will be
(i) same (ii) half
(iii) double (iv) zero



(e) The total energy possessed by a system of moving bodies
(i) is minimum in the start and maximum at the end
(ii) varies from point to point
(iii) is maximum in the start and minimum at the end
(iv) is constant at every instant

(f) Principle of transmissibility for free body diagrams is:
(i) It states that the force acting on the body is a sliding vector
(ii) It states that the force acting on the body is a rolling vector
(iii) It states that the force acting on the body is a wedging vector
(iv) It states that the force acting on the body is a unit vector

(g) The maximum frictional force which comes into play when a body just begins to slide over another surface is called
(i) dynamic friction (ii) sliding friction
(iii) limiting friction (iv) kinematic friction.

(h) The motion of a particle (distance in metres and time in seconds) is given by the equation $S = 2t^3 + 3t$, starting from t=0, to attain a velocity of 9m/s, the particle will have to travel a distance of
(i) 15 m (ii) 10 m (iii) 5 m (iv) zero

(i) A body of weight W is placed on an inclined plane. The angle made by the inclined plane with horizontal, when the body is on the point of moving down is called
(i) angle of inclination (ii) angle of repose
(iii) angle of friction (iv) angle of limiting friction

(j) When the car moves on road its wheel has
(i) Purely rotational motion (ii) Purely translational motion
(iii) Rotational and translational motion (iv) None of the above

Q.2 (a) A 5 m ladder weighing 250 N is placed against a smooth vertical wall with its lower end 3 m away from the wall as shown in fig-1. If the coefficient of friction between the ladder and the floor is 0.3, show that the ladder will remain in equilibrium in this position. **[7]**

(b) Block weighing 1000 N is resting on a horizontal surface. The coefficient of friction between the block and the horizontal surface is μ=0.2. A vertical cable attached to the block provides partial support as shown in fig-2. A man can pull horizontally with a force of 100 N. What will be the tension, T (in N) in the cable if the man is just able to move the block to the right? **[7]**

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Q.3 (a) A uniform wheel of 600 mm diameter, weighing 10KN rests against a rigid rectangular block of 150mm height as shown in fig-3. Find the least pull, through the centre of the wheel, required just to turn the wheel over the corner A of the block. Also find the reaction of the block. Take the entire surface to be smooth. [7]

(b) The mass of each ball is 200 grams, and connected by a cord. The length of the cord is 80 cm, and the width of the cord is 40 cm. What is the moment of inertia of the balls about the axis of rotation (Ignore cord's mass)? [7]

Q.4 (a) A beam 3m long weighing 400 N is suspended in a Horizontal position by two vertical strings, each of which can withstand a maximum tension of 350 N only as shown in fig-4. How for a body of 200N weight be placed on the beam. So that one of the strings may just break? [7]

(b) Smooth circular cylinder of radius 0.25 meter is lying in a triangular groove, one side of which makes 30° angle and the other 45° angle with the horizontal. Find the reactions at the surfaces of contact, if there is no friction and the cylinder weights 100 N. [7]

Q.5 A 8 m long simply supported beam with overhangs, rests on supports are 4 m apart. The left end overhanging is 3 m. The beam carries load of 20 kN and 10 kN on the left and the right ends respectively. Draw S.F.D & B.M.D. Locate point of contraflexure, if any. [14]

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Q.6 (a) Obtain the metric tensor for two dimensional plane in polar coordinates. [7]

(b) Show that any tensor of rank 2 can be expressed as sum of a symmetric and an antisymmetric tensors of rank 2. [7]

Q.7 (a) The angular velocity of the rigid body is defined by the vector: $W = w_1 i + w_2 j + w_3 k$, Obtain an expression for this angular velocity in terms of the Euler angles ϕ, ψ and Θ in the i, j, and k directions. [7]

(b) A car moving with a velocity of 10 m/s shows down in such a manner that the relation between velocity and time is given by: $= 10 - t^2 - \frac{t^3}{2}$, Find the distance travelled in two seconds, average velocity and average retardation of the car in these two seconds. [7]

Q.8 (a) A man weighing 750 N stands on the floor of a lift. Find the pressure exerted on the floor when (a) the lift moves upwards with an acceleration of 3 m/sec^2 and (b) the lift moves downwards with an acceleration of 3 m/sec^2. [7]

(b) A solid shaft transmits 200 kW of power at 600 rpm. Determine the suitable diameter of the shaft if the shear stress is not to exceed 70 MPa and total angle of twist is limited to 30 for 4m length of the shaft, Modulus of rigidity (G) =80 GPa. [7]

Q.9 Write short notes on ***any two*** of the following: **[7 x 2=14]**

(a) Symmetric and antisymmetric tensors

(b) Newton –Euler laws of rigid body motion

(c) Relation between load intensity, shear force and bending moment

(d) Gyroscopic motion

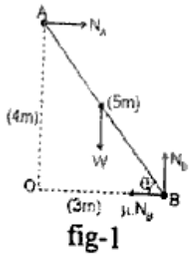


fig-1

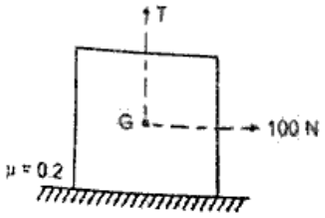


fig-2

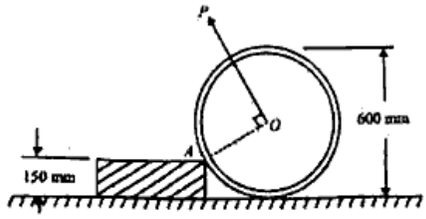


fig-3

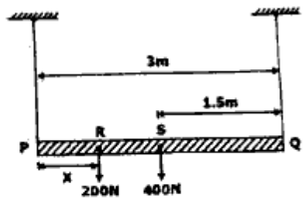


fig-4

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